

DFT calculation

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1 Destabilisation of MgH_2 by substitutional Ni and Co dopants: a first-principles study

Pure MgH_2 stores 7.6 % hydrogen by mass, but its formation enthalpy ($\Delta H_f \approx -75 \text{ kJ mol}^{-1} \text{ H}_2$, Bogdanović et al. [1]) is so negative – the hydride so stable – that release requires temperatures around 300 °C. A useful catalyst for this system is one that destabilises the hydride – makes ΔH less negative – so that release happens at lower temperature. Nickel in particular is known to improve the hydrogen sorption properties of MgH_2 [2]. To provide a first-principles rationale for the Ni- and Co-doped magnesium hydrides investigated experimentally in this work, we carried out density-functional theory (DFT) calculations of the hydride formation enthalpy. Within a single internally consistent setup we quantify whether substitutional Ni and Co at the Mg site destabilise the hydride at a fixed dopant concentration.

1.1 Methods

We compare three hydrogenation pathways on a common per- H_2 basis:



The two doped pathways use the same supercell topology – a $2 \times 2 \times 2$ replication of the rutile MgH_2 cell for the products and the matching $2 \times 2 \times 2$ hexagonal close-packed Mg cell for the reactants, with two of the sixteen metal sites occupied by the dopant $\text{X}=\text{Ni}, \text{Co}$, i.e. 12.5 atomic percent (at.%) of the metal atoms. The two doped pathways differ only in the chemical identity of the dopant, and each is referenced to the same pristine pathway, so the destabilisation $\Delta\Delta H$ relative to pure MgH_2 benefits from cancellation of supercell-size, k -point-sampling, and symmetry errors. Figure 1 shows the three relaxed product structures.

All calculations are performed with Quantum ESPRESSO [3]. The exchange-correlation functional is the Perdew–Burke–Ernzerhof (PBE) generalised-gradient approximation, augmented with Grimme-D3 dispersion [4]. Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (charge density). Pseudopotentials are of the projector augmented-wave (PAW) type for Mg and H and Vanderbilt ultrasoft pseudopotentials (USPP) for Ni and Co. Brillouin-zone sampling uses Γ -centred Monkhorst–Pack grids of $12 \times 12 \times 8$ for the Mg primitive, $8 \times 8 \times 12$ for rutile MgH_2 , $6 \times 6 \times 4$ for the Mg supercells, $4 \times 4 \times 6$ for the hydride supercells, and the Γ point for the isolated H_2 molecule. Metallic phases are described with cold smearing of width 0.01 Ry, while the isolated H_2 molecule uses fixed occupations. The self-consistent field (SCF) cycle is converged to 1×10^{-9} Ry using Anderson charge-density mixing. Cell parameters and atomic positions are relaxed simultaneously with a quasi-Newton scheme until the residual forces fall below 1×10^{-4} Ry a_0^{-1} . For the doped supercells no crystal symmetry is imposed during relaxation,

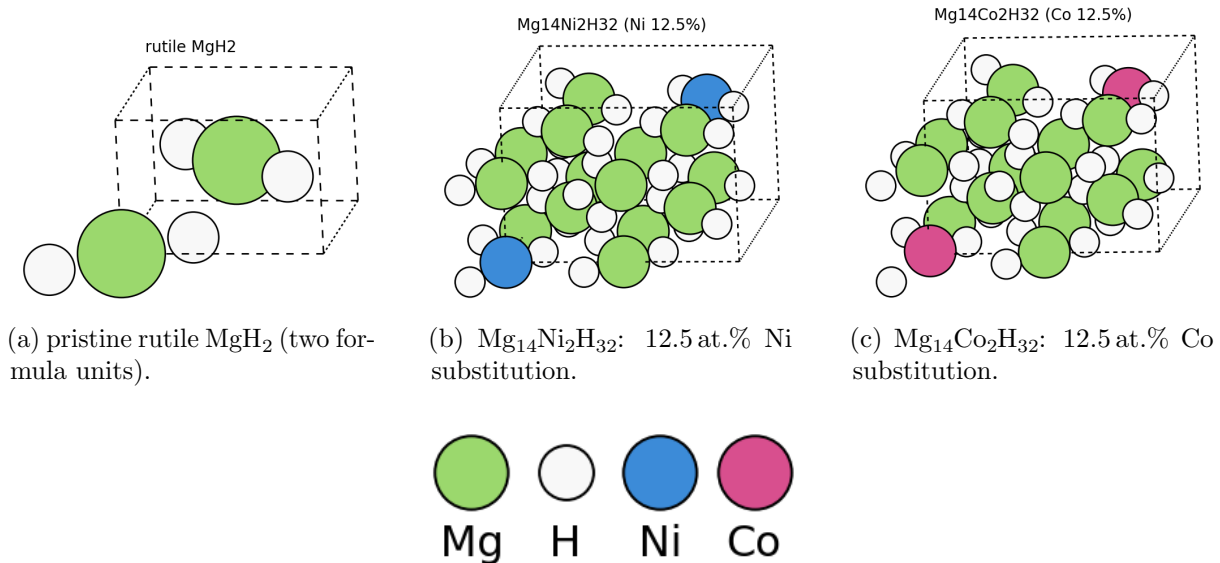


Figure 1: Relaxed product structures. The two transition-metal atoms occupy diagonally opposite Mg sites in the $2 \times 2 \times 2$ rutile supercell, maximising their separation to minimise dopant–dopant interaction.

so that the small unit-cell drift accumulated over the optimisation does not interfere with the symmetry analysis.

For each pathway the per- H_2 formation enthalpy is

$$\Delta H = \frac{1}{n_{\text{H}_2}} [E(\text{hydride cell}) - E(\text{metal cell}) - n_{\text{H}_2} E(\text{H}_2)] \times 1312.75 \text{ kJ mol}^{-1} \text{ Ry}^{-1}. \quad (1)$$

We use a non-spin-polarised treatment throughout. A spin-polarised check on $\text{Mg}_{14}\text{Ni}_2$ gave $0 \mu_{\text{B}}$ per Ni, so spin polarisation has no effect on the Ni pathway, and in test runs its influence on ΔH was below $1 \text{ kJ mol}^{-1} \text{ H}_2$. Cobalt carries a moment of $\sim 1.3 \mu_{\text{B}}$ per atom in the metallic reactant that is largely quenched by H bonding in the hydride; treating reactant and product on the same non-magnetic footing is expected to cancel most of the resulting bias in ΔH , and it also removes a spin-related convergence instability that otherwise prevented the SCF cycle from converging for $\text{Mg}_{14}\text{X}_2\text{H}_{32}$. The residual spin correction for the Co pathway has not been quantified and is the main uncertainty specific to that pathway, although given its magnitude relative to $\Delta\Delta H$ it is not expected to alter the qualitative ordering.

1.2 DFT results

All seven calculations completed successfully, with the SCF cycle converged at every step of the relaxation. The geometry optimisation reached $|\mathbf{F}| < 5 \times 10^{-4} \text{ Ry a}_0^{-1}$ on six structures; for $\text{Mg}_{14}\text{Ni}_2\text{H}_{32}$ the relaxation stalled at $|\mathbf{F}| = 5 \times 10^{-3} \text{ Ry a}_0^{-1}$, but its total energy was already converged to within one part in 10^6 , so the residual numerical uncertainty is $\lesssim 1 \text{ kJ mol}^{-1} \text{ H}_2$. The resulting formation enthalpies are summarised in Table 1.

The pure-Mg baseline ($\Delta H = -63.75 \text{ kJ mol}^{-1} \text{ H}_2$) sits about 11 kJ mol^{-1} above experiment ($-75 \text{ kJ mol}^{-1} \text{ H}_2$ [1]), reflecting the known PBE under-binding of hydrides [5]. The inter-pathway differences $\Delta\Delta H$ are differential quantities and benefit from cancellation of this systematic under-binding error, so they are meaningful even at the absolute level of accuracy delivered here.

Table 1 shows that both Ni and Co dopants at 12.5 at.% destabilise MgH_2 relative to the pristine pathway: both $\Delta\Delta H$ values are positive and exceed by an order of magnitude the residual numerical uncertainty of the calculation. Each dopant therefore makes the hydride

Pathway	ΔH / kJ mol ⁻¹ H ₂	$\Delta\Delta H$ vs. pure
pure Mg	-63.75	—
Ni 12.5% Mg ₁₄ Ni ₂	-48.80	14.95
Co 12.5% Mg ₁₄ Co ₂	-55.24	8.51

Table 1: Formation enthalpies of magnesium hydride per absorbed H₂ for the three pathways. Less-negative ΔH means the hydride is less stable, i.e. easier to dehydrogenate – the desired catalyst effect.

easier to dehydrogenate than pure MgH₂. This is consistent with the long-standing experimental finding that Ni-containing magnesium hydrides are less stable than pure MgH₂: the dedicated ternary phase Mg₂NiH₄ desorbs with $\Delta H \approx -64.4$ kJ mol⁻¹ H₂ [6], some 10 kJ mol⁻¹ less stable than MgH₂. We emphasise that this is a comparison of trend only: our Mg₁₄Ni₂H₃₂ models dilute (12.5 at.%) substitution within the rutile MgH₂ lattice, which is a distinct system from the ordered, Ni-rich Mg₂NiH₄ phase, so the absolute enthalpies are not expected to coincide.

In this thermodynamic sense a catalyst weakens the H–metal bond just enough to lower the release temperature without sacrificing storage capacity, and substitutional Ni or Co does exactly that. These calculations thus provide a thermodynamic rationale for the improved dehydrogenation observed for the Ni- and Co-doped samples in this work.

References

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